

# Ethan Phan

New York, NY | 631-576-6328 | ethanlphan8@gmail.com | linkedin.com/in/ethanlphan

## EDUCATION

### University at Buffalo, The State University of New York

August 2021 – December 2024

*Bachelor of Science in Computer Science*

*GPA: 4.000*

- Honors: Pride of New York Scholarship, Dean's List
- Relevant Courses: Software Engineering, Algorithms and Complexity, Web Applications, Modern Networking Concepts, UNIX Systems Administration, Intro to Artificial Intelligence, Intro to Machine Learning

## EXPERIENCE

### Research Assistant (Student Programmer)

August 2024 – Present

*Network Science Institute, Northeastern University*

*Remote*

- Utilized Natural Language Processing (NLP) techniques to analyze data, investigating the causes of misinformation spread among older adults on Facebook.
- Developed a Python and Selenium system integrated with Google API to dynamically scrape data from 100+ Facebook profiles, enabling scalable data collection and adaptation to varying post volumes per account.
- Designed and implemented API endpoints with Flask for efficient storage of scraped data into a MySQL database, enhancing data accessibility and scalability for ongoing research.

### Teaching Assistant

August 2022 – December 2024

*School of Engineering and Applied Sciences, University at Buffalo*

*Buffalo, NY*

- Instructed 600+ students on topics including Data Structures, Societal Computing, and Discrete Mathematics.
- Led four 1-hour recitations, addressed over 50 student questions, and held 5+ hours of office hours each week.
- Evaluated assignments and delivered feedback to enhance student understanding of material and performance.
- Identified course improvement areas and provided actionable recommendations to enhance structure and engagement.

### Technology Intern

June 2023 – August 2023

*M&T Bank*

*Buffalo, NY*

- Streamlined deployment by integrating Docker into GitLab pipelines for 200+ .NET web applications.
- Led project design, enhancing performance and sustainability, resulting in \$500,000 in annual savings for the bank.
- Gained expertise in Docker, YAML, Angular, C#, and .NET by building a robust CI/CD pipeline and web application in an agile environment.
- Recognized and awarded for demonstrating exceptional communication and teamwork, ensuring project success.

## PROJECTS

### Convolutional Neural Network | *Python, PyTorch, NumPy*

- Constructed a Convolutional Neural Network (CNN) in PyTorch to classify handwritten digit images, achieving 98.27% accuracy.
- Enhanced model performance using Stochastic Gradient Descent (SGD) and compared results across batch sizes to evaluate training efficiency.

### Productivity Management Platform | *Angular, TypeScript, SCSS, Express.js, MongoDB*

- Engineered a penguin-themed productivity web application that incentivized task competition and study sessions.
- Created a user-friendly front end with 25+ reusable components and built a RESTful API for user authentication and data management.

### Penguin Reinforcement Learning | *Python*

- Drafted and established a reinforcement learning framework using SARSA, Q-Learning, and Double Q-Learning for environmental navigation insights.
- Optimized algorithms through hyperparameter tuning across 10,000+ episodes and visualized results with pandas, NumPy, and Matplotlib.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, C#, Java, JavaScript, HTML/CSS

**Web Development:** Angular, React, SCSS, Tailwind CSS, TypeScript, Express, .NET, Flask

**Technologies & Tools:** UNIX, Git, GitHub, GitLab, Node.js, MySQL, MongoDB, Docker

**Product Management Tools:** Trello, Jira, Confluence